

2024
4798/5

FACULTY OF SCIENCE

B.Sc. (I Semester) Examination

STATISTICS

Paper I

(Descriptive Statistics and Probability)

Time : 3 Hours]

[Max. Marks : 80

Section A – (Marks: $8 \times 4 = 32$)

(Short Answer Questions)

Answer any eight of the following:

1. Explain method of frequency distribution for an ungrouped data.

2. Write a short note on skewness and kurtosis.

3. Calculate mean, median, mode for the following data.

Class Interval	0-3	3-5	5-7	7-9	9-11
frequency	4	5	3	6	2

4. Define the following terms with an example:

i) Sample space

ii) Mutually exclusive

5. State two models of Boole's inequalities in probability.

6. Two dice are thrown simultaneously, find the probability of getting:

(a) same number occurrence on the two faces

(b) The sum neither 7 nor 10

7. Define discrete and continuous random variable with an example.

8. Explain the concept of transformation of one-dimensional random variable.

9. A random variable X has the following probability function:

X :	0	1	2	3	4	5	6	7
P(x) :	0	m	2m	2m	3m	m ²	2m ²	7m ² +m

(i) Find m, (ii) Evaluate $P(X < 3)$, $P(X \geq 3)$ and $P(0 < X < 4)$

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- 10 Define the term 'Mathematical expectation' and give one example with discrete case only also state its addition and multiplication theorem.
- 11 Define probability generating function of a random variable X also state its properties.
12. A random variable X has the following probability function:

X	2	4	6	8	10
$P(x)$	$1/3$	$1/5$	$1/7$	$1/9$	$1/11$

Determine $E(X)$ and $V(X)$

Section B – (Marks: $4 \times 12 = 48$)

(Essay Type Questions)

Answer all questions.

13. (a) Explain primary and secondary data collection methods with an illustrations.
Or
~~(b)~~ Define central and non central moments and derive the relation between them.
14. (a) State and prove addition theorem of probability for 'n' events..
Or
~~(b)~~ What is Conditional Probability? State and prove Baye's theorem of probability.
15. (a) Explain probability mass function and probability density function with illustrations.
Or
(b) Define Distribution function of a random variable X and prove all its properties.
16. (a) Define moment generating function of random variable X , how to obtain moments using moment generating function.
Or
~~(b)~~ State and prove Chebychev's in equality.
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FACULTY OF SCIENCE

B.Sc. (II Semester) Examination

STATISTICS

Paper II

(Probability Distributions)

(New)

Time: 3 Hours]

[Max. Marks: 80

Section A – (Marks: $8 \times 4 = 32$)

(Short Answer Questions)

Answer any eight questions.

1. Define Binomial distribution state its mean and variance also write its real life applications.
2. Define geometric distribution also derive its lack of memory property.
3. A coffee connoisseur claims that he can distinguish between a cup of instant coffee and a cup of percolator coffee 75% of the time. It is agreed that his claim will be accepted if he correctly identifies at least 5 of the 6 cups. Find his chances of having the claim (i) accepted (ii) rejected, when he does have the ability he claims.
4. Define hyper geometric distribution with an example.
5. Define rectangular distribution also derive its mean and variance.
6. The Nizamabad municipal corporation had installed 10000 LED light bulbs in the different parts of the city. An average life of these light bulbs is 1000 burning hours with standard deviation of 200 hours. Find the number of light bulbs that are expected (i) to fail in the first 800 burning hours and those are expected (ii) to burn between 800 and 1300 burning hours. You may assume that the life data of the light bulbs follow a normal distribution.
7. Derive the moment generating function of single parameter gamma distribution.
8. Define second kind of beta distribution also state any two properties of it.
9. In a metro railway station, it is observed that the mean inter arrival time of trains is 20 minutes. Assuming inter arrival time follows exponential distribution. Find the probability that the inter arrival time is more than 25 minutes.
10. Define the following terms with an example.
(i) Population (ii) Sample (iii) Parameter (iv) Statistic

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11. Explain about sampling distribution and standard error.
12. Define any two exact sampling distributions of your own choice also state properties of anyone of them.

Section B – (Marks: $4 \times 12 = 48$)

(Essay Type Questions)

Answer **all** questions.

13. (a) Define Poisson distribution obtain its m.g.f. and hence find mean and variance of Poisson distribution.
- Or
- (b) Define Negative binomial distribution. Show that mean is less than variance for negative Binomial distribution.
14. (a) Define Normal distribution. Explain the chief characteristics of normal distribution also discuss its additive property
- Or
- (b) Show that in normal distribution all odd moments $\mu_{2n+1} = 0$ and even order moments $\mu_{2n} = 1.3.5. \dots (2n-1)\sigma^{2n}$
15. (a) Define exponential distribution. Derive the constants of exponential distribution.
- Or
- (b) Define Cauchy distribution and derive its characteristic function.
16. (a) Derive the Chi-square distribution also mention its applications.
- Or
- (b) Derive the relation between any two distributions of t, F and χ^2 with your own choice.
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2025
Sem - IV

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FACULTY OF SCIENCE

B.Sc. (IV Semester) Examination

STATISTICS

Paper IV

(Statistical Inference)

Time: 3 Hours]

[Max. Marks: 80

Section A – (Marks: $8 \times 4 = 32$)

(Short Answer Questions)

Answer any **eight** questions.

- ① Define level of significance with an example also give common level of significance in testing of hypothesis.
- ② Discuss the concept of randomized and non-randomized test function.
3. Let θ be the probability that a coin will fall head in a single toss in order to test $H_0: \theta = \frac{1}{2}$ against $H_1: \theta = \frac{3}{4}$. The coin is tossed 5 times and H_0 is rejected if more than 3 heads are obtained. Find the probabilities of type I and type II errors and power of the test.
- ④ Explain the test procedure for testing the difference of two proportions.
5. Define Order Statistics also state its distributions.
6. In a sample of 1000 people in Maharashtra, 540 are rice eaters and rest is wheat eaters. Can we assume both rice and wheat eaters are equally proportion in this state at 5% level of significance.
- ⑦ Write a short note on t-test and also give its test procedure of single sample mean.
- ⑧ What is meant by contingency table? Explain the procedure of 2 X 2 contingency table.
9. The number of automobile accidents per week in a certain community was as follows. 14, 10, 22, 4, 16, 12, 17, 8, 11, 6, are these frequencies in agreements with the belief that accident conditions were the same during this 10-week period.
(χ^2 -table value at 5% L.O.S and 9 d.f = 16.91)
10. Explain about different measurement scales in non-parametric tests.
11. Explain the importance of central limit theorem in non-parametric tests.
12. Test the randomness of the sample: SSRSSSRSSSSRSRRR

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Section B – (Marks: $4 \times 12 = 48$)

(Essay Type Questions)

Answer **all** questions.

13. (a) Define the following terms with an example:

- (i) Null hypothesis and alternative hypothesis
- (ii) Type-I error and Type-II error
- (iii) One tailed and two tailed tests

Or

(b) What is BCR? State and prove Neyman-Pearson's fundamental lemma for finding BCR.

14. (a) Explain the test procedure for testing the difference of two standard deviations.

Or

(b) Explain the test procedure for testing the significance of two sample population correlation coefficient.

15. (a) Define chi-square test also state its applications. Explain chi-square test for goodness of fit.

Or

(b) Describe the t-test procedure for testing the significance of difference between two samples means in case of independent and related samples.

16. (a) Discuss about non-parametric methods. Explain merits and demerits of non-parametric methods.

Or

(b) Explain sign test and Wilcoxon-signed rank test in case of single and paired samples.

2024

5075/4

FACULTY OF SCIENCE
B.Sc. (III Semester) Examination
BIO-STATISTICS
 (Skill Enhancement Course-2)
 (New)

Time: 2 Hours]

[Max.Marks: 40

Section A- (Marks: $4 \times 4 = 16$)

(Short Answer Questions)

1 Answer any **four** of the following:

- (a) Explain various data collection methods.
- (b) Define correlation with an example also mention limits of correlation coefficient.
- (c) Calculate mean, median and mode of 35, 40, 20, 25, 30, 40, 50
- (d) Define the term probability with an example.
- (e) Define normal distribution and state its properties.
- (f) Two dice are thrown; Find the probability of the same number occurrence on two dice.

Section B- (Marks: $2 \times 12 = 24$)

(Essay Type Questions)

2. (a) Explain the construction procedure of histogram also discuss the concept of skewness.
 Or
 (b) Calculate coefficient of correlation for the following data.

X	12	14	13	10	8	11	7
Y	7	8	12	14	13	10	9

3. (a) Define the following terms with an example.
- i) Sample space
 - ii) Mutually exclusive events
 - iii) Binomial and Poisson distribution
 - iv) Two types of hypothesis
 - v) Type-I and Type-II errors
 - vi) Level of significance

Or

(b) In a Mendelian experiment on breeding, four types of plants are expected to occur in the proportion of 9: 3: 3: 1. The observed frequencies are: 891 round and yellow, 316 wrinkled and yellow, 290 round and green and 119 wrinkled and green. Total 1616. Find the Chi-square test value and examine the correspondence between the theory and experiment. The tabular value of Chi-square for three degree of freedom at 5% level of significance is 7.81.

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